LIGHT:
A FAIRY TALE

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LIGHT: A FAIRY TALE

King John, Fourteen Portraits, and a Sorceress

ONCE upon a time, there was a young prince named John who had just been crowned King of Amrath. The new king was an orphan, raised by maids and butlers and looked after by his steward, Griselda, who had managed the kingdom until John came of age. Taking the throne was terrible news, both for the citizens of Amrath, who just wanted to go about plowing their fields and treating their goats' hoof rot. And for John himself, who only wanted, in an abstract way, to slay dragons and rescue princesses. In fact, the idea of being king was so terrible to him that John avoided the throne room as much as possible. This was partly because there was nothing fun to do there, and partly because of the fourteen enormous portraits of his forebears looming down at him from high up on the walls.

Today, however, the portraits were not on his mind.

John stood on the parapet overlooking the crowd in the square below.

Most of them were screaming, but not all of them were demanding his execution. Some of them wanted him thrown in

prison. Others were focused on throwing rotten vegetables at him.

The crown on John's head was cold, the metal biting into his scalp. Someone down below had quite an arm, having hurled a potato which smacked into his jaw. The new king stumbled and reached up to keep his crown from tumbling. Standing next to him, Griselda shifted her weight to the other foot.

John wished he hadn't spent the week before his coronation reading gossip about the nobles. He wished he hadn't spent years before becoming king reading adventure stories about young princes who saved kingdoms from dragons. If only there were a dragon around. That was why no one liked him. He needed to kill a dragon. Maybe he could find one. Or declare war. He'd tried that, though, and the neighboring kingdom had ignored him, and the troops had refused to march.

He took a deep breath. He was now King John XV and as he looked down at his subjects, he saw their faces contorted with anger and contempt. He looked at Griselda, who smiled encouragingly but without a lot of confidence. John considered giving up. But then he remembered the story of Prince Henry III, who rode dragons and burnt his enemies to a crisp while rescuing maidens and increasing the economic production of the kingdom. He thought of King John XII, his great-grandmother (yes, his own great-grandmother was named after a man; the naming tradition really was unfortunate for the women of the family). She had ridden into battle with the wild men in the north, driving them back. She hadn't given up,

even when she'd been captured and beheaded. He wished he could have met her. She could have told him what to do now.

John readjusted his crown, squared his shoulders, and approached the end of the parapet, dodging a carrot as it sailed past. He planted his feet on the golden circle where his father had stood to announce his birth, only days before both he and John's mother had become ill and died. John cleared his throat, and the sound, magnified as if by magic, echoed off the perfectly designed walls of the square. The furious shouting quieted to an angry muttering.

He opened his mouth, not knowing what he was going to say, but just then the sun came out from behind a cloud and shone straight in his face, blinding him. The crowd gasped, and a hush settled over the square. The new king stood bathed in a ray of sunlight, not daring to believe his luck. John struggled not to blink, but to stare stoically into the sun with what he hoped was a kingly expression on his face.

"Since the death of my parents, dark times have fallen on our land," he said. "Steward Griselda has done an admirable job, but a kingdom needs a king." That last line came from The Adventures of King Edward the Relentless. "I will be a light in the darkness. I will illuminate the darkest corners of our kingdom." He could feel the sunbeam starting to fade. Time to wrap it up. "Light is coming to Amrath." He turned and walked away from the parapet just as the cloud moved back over the sun. Behind him, the crowd explode into talk. Griselda's gaze landed on him. He grinned.

Then he tripped on the carpeting. His crown tumbled off his head, but he caught it, tucked it under his arm, and

looked around to see if anyone had noticed. The soldiers lining the walls stared straight ahead. Someone laughed, but it was quickly turned into a cough.

The next few weeks did not go as King John had hoped. He commissioned several statues of himself to be put up around the kingdom, got himself a sword-fighting teacher, and rode out into the woods looking for dragons and was nearly mauled by a boar. The peasants, briefly impressed by what had appeared to be an act of the divine during his speech, rapidly returned to their grumbling, and the square outside the palace collected protesters again.

One evening, feeling morose and deciding that the most heroic place to be morose was the throne room, King John XV went and lay prone on the carpeting, looking up at his heroic, judgmental ancestors.

"I know I'm doing a terrible job," he said defensively.

To his surprise, someone answered. "Yes, you are."

King John had heard of people hallucinating conversations with portraits or statues or even things like bedside tables or chamber pots—you'd be surprised how good the advice of a hallucinated chamber pot can be—but he'd never experienced it himself. Still, at this point John was up for anything.

"Yes, oh glorious ancestors! What must I do?" This was what people said in the sorts of books he liked to read.

Then he saw the woman standing underneath the portraits.

"Oh, I'm sorry, I didn't see you there." On second thought, that was probably too apologetic a thing for a king to have said, and John wished he hadn't said it, but it was too late now. At least it was polite. And it was entirely possible that she was a hallucination as well. It was probably best to be polite to hallucinations.

The—possibly hallucinated—woman was old, with a hunched back and short, thick fingers. She clutched a lumpy grey candle in one hand.

"I heard of your speech, young king," she said. Her voice was soft now, barely above a whisper. "I can teach you to be a light in the darkness, if you wish."

John didn't think she could teach him any of things he had in mind. Like troll-wrestling. On the other hand, this was often the way things started in stories, so he went along with it.

"What do I have to do?" After all, maybe she was a powerful sorceress. She looked like a powerful sorceress. Maybe it was better to check. "Are you a powerful sorceress?" he asked quickly.

She blinked. "Maybe." For a moment, her image wavered, and John thought he saw a young woman with long, white hair and pointed ears, glowing with a bright purple and white aura. But it must have been a trick of the light because a moment later the vision was gone.

The old woman cleared her throat and held up the candle. "Take this candle. Go to the top of the highest tower and light it. You will learn all you need to learn."

John reached out for the candle, but just as his fingers touched it, she pulled back. "Take care, though. Do not let the candle burn all the way to the end."

John nodded solemnly, but immediately decided he would let the candle burn all the way to the end, just to see what would happen.

"Thank you, great sorceress," he said. When he looked up from the candle, she was gone, and he was alone in the throne room. Except, of course, for the horrible portraits on the walls.

Immediately, King John climbed the rickety, winding stairs—all three hundred and seven of them—to the top of the tallest tower. Disconcertingly, the tower tended to sway and, as it was a windy evening, there was more than a slight wobble. Luckily, despite his many failings, John was at least totally unafraid of heights. He'd always hoped it would come in handy for dragon riding someday. If he ever found a dragon.

For ten or fifteen minutes, King John stood lost in thought at the top of the tower, imagining what it would be like to ride a dragon, but then he remembered why he was there. He took out the candle and realized he hadn't brought any matches. (Matches had been invented by King John III, who had also invented heated carriage seats and GPS. To be fair, GPS was simply Gregory Pontius Samson, a local mapmaker whom King John III had employed to lead him around town because he was always getting lost. But still. The matches were a useful invention.)

Annoyed with himself for forgetting the matches, John trudged back down the three hundred and seven stairs, found a packet in a kitchen drawer, got distracted taste-testing mince pies for forty-five minutes, drank several pints of ale with the disconsolate cook who had recently broken off his engagement to the baker, took a two-hour nap, then returned to the top of the tower.

By now it was quite late.

Looking down, he noticed a large crowd still occupied the square. The square was called Proclamation Square, but John had recently come to think of it as Protestor Square. The protestors were waving torches, and John could hear faint shouts. He was going to have to do something about the unrest soon. He'd heard what happened to kings when peasants got too angry.

Well, he had this candle now, given to him by a powerful sorceress. That was something. John took a deep breath. His stomach, full of mince pies and ale, strained and groaned.

King John set the candle upright in the center of the tower, squishing it around until its waxy bottom was well-stuck to the sandstone. Then he struck a match and held the flame to the wick. The fire changed color as the wick caught, burning with a diamond-like brilliance. John felt as if the light shone straight through him; he could feel it illuminating his bones with its white, shining light. He looked around and saw that all the color had gone from the world. It shone in black and white with every shade of grey shimmering in between.

He could see straight through the tower beneath him, down through every step, down to the kitchens beneath. He

could see the bones and the eyeballs of the cook from above, could see his heart and his stomach—also full of mince pies. Beneath the cook were more flag stones, and storerooms, and a thief sneaking through a culvert John hadn't known existed. He turned to see the rest of the castle, and there was the throne room with the portraits on the walls. There was something behind them that he couldn't quite make out. John made a mental note to check later.

He could see every maid and butler and swordsman in the entire castle. He could see the femurs of the protestors down in the square.

For several hours, John simply sat and gazed at the strange view he had on the world. It didn't seem to be wearing off, and the candle didn't seem to be burning down at all, which surprised him. Then, he began to wonder. While entertaining, he wasn't sure what he was supposed to be learning from this. How was this supposed to help him be a better king? He turned to look at the candle.

"Oh, great candle, help me to be a light for my people, teach me to illuminate my kingdom."

There was a faint pop and suddenly John was speeding through the air. If he'd had time to think about it, he'd have noticed that he wasn't feeling the wind in his hair because he didn't have any hair. But he didn't. He was moving too quickly.

He sped in a straight line, hurtling over the rooftops of the town. He saw he was approaching a tree and tried to close his eyes. That was when he noticed that he didn't have eyes. He still didn't notice the hair, though. He tried to flail his arms and legs, to turn or slow himself, and discovered he didn't have

those, either. For some reason that came as a relief. At least they wouldn't be injured when he crashed into the tree.

He collided with a single, large maple leaf and felt himself divide. Part of himself absorbed into the leaf, and part of himself bounced off. His mind stayed with the part that bounced, and he felt different. Less. But just as fast. He hit another leaf and bounced off that, too, this time losing nothing. Then he hit the trunk, and that was a different story. The last of himself soaked into the trunk, and he was gone.

There was another faint pop, and John found himself standing at the top of the tower again. He immediately fell over.

When at last he was able to push himself up to a seat, and, to his relief, found that he again had arms and legs and eyeballs (he still hadn't noticed the hair, although that was back, too). He saw that the candle had gone out, and that it was a quarter of an inch shorter.

"What in the world was that?" John mused aloud, not expecting an answer.

"You asked to be light," a sonorous voice proclaimed.

John whirled around and saw not one but five ghosts floating above the other side of the tower. One of them was trying to hide behind the others, but as they were all transparent it wasn't working. In fact, John found himself staring at that one in particular. He had a shock of pearly white hair.

"That's Snell. Don't mind him," one of the others said.

"Er," said John, and left it at that.

The six of them stood staring at one another. One of the ghosts coughed politely. Two of them swung their arms. Snell

looked at his shoes, and the one directly in front of him attempted to whistle but failed utterly.

"Excuse me," John said. "My name is John. King John the Fifteenth. Pleased to meet you; welcome to Amrath."

The ghosts looked relieved, and the one closest smiled, held out his hand, then withdrew it sheepishly. "You'll have to excuse us. We're not used to being transparent." He cleared his throat. "Or incorporeal."

John waved a hand as if it were a trivial matter, and the ghost, who had a crazy cloud of white hair floating around his head, continued.

"We are the five light mages. Or we were. The witch has trapped us here in this candle, forcing us to do her bidding."

"Oh, I'm so sorry to hear that," John said, feeling bad and wondering if it was still ethical for him to ask for their help in improving his kingdom. He'd just wanted some magical help; he didn't want to enslave anybody. "Is there any way for me to free you?"

"If you burn the candle all the way to its end, we will be freed."

Well, he'd been planning on doing that anyway, so that was convenient. "Well, of course; I'll do that immediately then, shall I?" He could figure out something else for his kingdom. John moved to light the candle, and all five ghosts shot forward in alarm, their transparent arms raised.

"Wait!"

"What is it?" John asked.

The ghost with the crazy hair continued, "This candle contains the five of us and two dark secrets. If you burn it, you will let them out, too."

"Well, that can't do that much harm, can it?"

"It will kill you."

"Oh." Now John felt really uncomfortable. There were five of them imprisoned. If he didn't let them out, they would all die in there, he assumed. But if he let them out, then he would die. Was his life worth the life of five mages? He was a king, that had to count for something, but he wasn't a very good king, and to be honest he didn't think he'd amount to much. The five light mages had probably done more for the world. But he didn't want to die, and that outweighed most other arguments in his mind.

He thought of King John V, who had sacrificed himself to save the small village of Oenwaith, which made John feel guilty. Then it occurred to him that someone else could light the candle, and then he felt really guilty and tried to pretend to himself that he hadn't thought of it.

"It's all right," the ghost with the crazy hair said. "You don't have to die. We can teach you to become a light mage, too. Then you can light the candle safely and free us."

John perked up at this. Becoming a powerful mage was exactly the kind of thing he'd had in mind when the sorceress had given him the candle. "Great!"

"It won't be easy," a dapper ghost behind the crazy-haired one said.

"Oh, that's fine," John said, smiling.

"No, really. It's going to be really hard."

"Oh, I'm sure it'll be ok."

"No, no. It's very dangerous." The other ghosts nodded. Even Snell.

"Awesome. I'm in." John grinned widely.

The ghosts glanced at each other.

"Very well," the crazy-haired one said. "To be a mage you must control the light. To control it, you must understand it. I will be your first teacher. My name is Albert Einstein, and I will teach you to ride the light."

Albert Einstein and the Speed of Light

John's grin became even wider, and his mouth opened several inches. He looked at the other ghosts to make sure they weren't joking. They all looked deadly serious. "Can we start now?" John asked.

Einstein glanced at the others. They shrugged. "All right. Let's begin."

John punched the air and jumped a few feet, then tried to appear as if he was taking things seriously.

The ghost of Einstein moved forward, reaching a misty hand towards the candle. A spark leapt from his fingers to the wick and light flared. Einstein lifted his hands, and time stopped.

All sound had ceased. Only John and Einstein could move. The ghosts hung like frozen clouds behind them. Between Einstein and John was a glowing yellow orb.

"This is a photon," Einstein said. "A particle of light."
"Awesome. How do I ride it?" John asked.

"By holding on tightly. I've stopped time, so you can get on. Otherwise, it moves far too fast. I'll float along next to you as you go."

Following Einstein's instructions, John climbed aboard the glowing particle. It was bright white and pulsed with energy.

"All right," Einstein said. "Have you got a good hold?"

John wasn't sure what he meant, because there was literally nothing to hold on to, but he didn't want to be a difficult student, so he nodded.

"Good. All right, I'll start time again. But slowly. Otherwise, you'd be on the other side of the kingdom in about a hundredth of a second."

"A hundred seconds?"

"No. One second chopped into a hundred pieces. Take one of those."

"Wow."

"Yes."

John was about to say something else, but Einstein clapped his hands together and the beam of light shot off into the night, with John clinging to it.

It was much smoother than riding a horse, strangely. It didn't jostle or turn or speed up or slow down at all. In fact, once John got used to it, it was a fairly boring ride. Einstein came up next to him.

"Seems like you're getting the hang of it."

"How do I make it go faster?"

"You can't."

"Turn?"

"You can't."

"This is it?"

They were fast approaching the ground.

"No, this is only the beginning."

John felt the photon impact the ground with a shudder, but the light only dimmed, and a second later they were shooting away in a slightly different direction.

"I thought you said we couldn't turn!" John yelled.

"You can't turn light, no, but it does bounce off things quite well."

"Why is it dimmer?"

"Some of it was absorbed into the ground."

"I'm still going just as fast, though."

"Indeed. You've lost energy, but you're not any slower. Just dimmer. And your color is different."

John hadn't noticed, but it was true.

He crashed into a boulder, and the light winked out. Gasping, he found himself again at the top of the tower.

The ghosts were still standing where they'd left them.

"I hope this doesn't get boring for you," John said. "You can go down to the kitchens, or, er, something. If you want."

"You were gone less than a second. We don't get bored that easily," the dapper one said.

"Oh, right."

Einstein reappeared.

"All right, not bad for your first try. Now it's time for your first challenge. To move on to your next teacher, you must turn your photon pink. You must then successfully return to the top of the tower, still astride your light beam."

"But . . . that was just my first try."

"So?"

"Don't you have more to teach me?"

"You'll find that there is actually very little to teach. The difficult part is not knowing but doing."

"Do I get any more practice first?"

"No."

"Are you going to give me any hints?"

"No."

"Pointers?"

"No."

"Will you slow time down for me again?"

"I'm not here to make it easy for you. Being a light mage is difficult, and most of it you have to do for yourself."

John didn't think this was how training was supposed to work, but it didn't seem like he was going to change Einstein's mind. He still wanted to be a light mage, so he thought hard about the challenge. Normally he would have taken a good long nap before trying to think this hard, but, for the first time he could remember, he didn't want to waste time. He didn't want to distract himself by reading an adventurous tale about dragons, either. He wanted to think.

So, John sat down on the cold flagstones and thought. How was he going to turn the light pink? Well, he'd turned it brown. That was pretty close. But why did it change color? It lost something when it hit the ground. Whatever that was had turned it brown. Was it something about how he'd hit the ground?

"Remember that light is what you use to see," Snell whispered. The others turned to glare at him, and he blushed.

Why did that matter? He used light to see, yes. Then he started to put it together. If he used light to see, then if he saw something pink, he must be seeing pink light! To turn the light pink, all he had to do was bounce off something pink. The rose garden was not far away. There were pink roses there.

He stood up in excitement and looked out over the ramparts. There were the roses. Now all he had to do was figure out how to get back. But light always went straight. The only way to turn was to bounce off something. The other things he'd sort of glanced off as he'd hit them. That gave him an idea.

Without stopping to second guess himself, John grabbed the candle, pointed it at the rose, and lit the wick.

The photon appeared, picking John up as it went past him. Time slowed down as he pulled himself up onto it and they shot towards the rose. There was no controlling or changing it now; all John could do was watch. They hit the rose dead on, exactly as John had hoped; the light beneath him dimmed and glowed pink, and they bounced straight back towards the tower.

The candle still hung in midair where John had let go of it. It hadn't even had time to fall yet. They crashed straight back into it, the light absorbing into the candle's dark surface, and John fell to the ground.

"Well done," Einstein said, sounding surprised and impressed.

"You slowed time for me," John said.

"I most certainly didn't."

"But it went so slow," John said, confused.

Einstein's eyebrows lifted into his white hair. "You must have slowed it yourself." He paused and looked at John more carefully. "Perhaps you are more suited to this than I realized."

No one had ever told John he was well-suited to something. Ever. It was an amazing feeling, and John wasn't sure how to take it, so he changed the subject.

"So, what's next then?"

The other ghosts, still looking at John with curiosity, stepped aside to reveal Snell.

Snell and the Law of Refraction

Snell looked so shy and uncomfortable. Wishing to put him at ease, John chose his words carefully. "Will you tell me more about how to ride the light?" he asked gently.

"No," Snell said, and his voice, though quiet, was sure. "I will teach you to *be* the light."

Without making eye contact, Snell moved forward and gestured to the candle. "Focus your attention on the wick."

John stared at the wick.

"No," Snell said, his annoyance making him quieter. "Focus on the wick until nothing exists but the wick."

"And me, right?"

"No. Just the wick."

Again, John felt he'd been given instructions that made no sense. But he would try to follow them. He stared at the wick. Other things, like thoughts and sounds and his own breath kept distracting him, but he focused on the wick. Suddenly, it

ignited. Immediately, John shot through the air, crashing into things, losing energy as he did so, and then he was gone. And then he was back on the tower.

"You were supposed to stop time," Snell said.

"I've only done it the once," John grumbled.

"Yes. So clearly you know how to do it."

John wanted to complain that Snell hadn't told him to stop time, nor had anyone ever told him how to do it, but again, he didn't want to be a bad student.

"Ok. Sorry. I'll try again."

The second time was quicker. He focused on the wick, letting everything else drop away, and suddenly he was a beam of light, shooting out into the darkness, and then he was nothing, and then he was back at the tower.

"Sorry," John said. "It was too fast."

"Again," Snell said.

"Sorry," John said. "I wasn't ready for it."

"Again," Snell said.

"Sorry," John said. "I forgot that time."

He wasn't even sure what he was supposed to be doing. All he knew was that he wanted to see what happened when he was the light.

The next time the candle lit, everything froze.

"Finally," John tried to say, but found that he couldn't talk. He was a glowing orb.

"Finally," Snell said. "Now, notice that you are white now, but that you are actually made up of more than that."

John looked at himself, and realized that, yes, he was in fact all colors.

"White light is really all the colors combined."

Ah, John thought, that explained the rose.

"When you hit something, some of those colors are absorbed, the other colors bounce off. Like in the rose. Now, go into the ocean."

John would have protested, but, as a photon, he had no way to protest. So, he pointed himself towards the ocean and then he started time again. Or he tried to. It didn't work. "Start!" he thought to himself. "Go!"

He willed himself forward, straining with everything he had, but nothing worked.

"This is impossible," he thought. Time started again, and he zipped off. An instant later he hit the surface of the water, and something strange happened.

At first, he thought time had slowed down again, but it hadn't. Everything around him was still going the same speed it had been before, but he had slowed down. He thought Einstein had said he couldn't change speed. The other strange thing that happened was that he turned. Deeper into the water.

Then he hit a turtle; the last of his energy was absorbed, and he was back on top of the tower, corporeal once again.

"All right," Snell said.

"What was that? I thought you said I couldn't change speed," he said to Einstein.

"I said you couldn't go any faster. It can't go faster. But it can slow down."

"Hey," Snell said. "You had your turn. You could have explained it then."

Einstein lifted his hands and floated off. Snell addressed John.

"Light goes more slowly through some things than others."

"Like what?"

"Like glass, and water. Diamonds. It's fastest when going through nothing. Almost as fast when going through air."

"Ok, but how do you explain my turn?" John wondered if this was something that only he could do. That would be super cool.

"It's very simple," Snell said, dashing John's hopes. "Imagine you're sliding down a ramp headfirst."

John wondered where that had come from, but he did as he was told.

"Now, imagine the right side of the ramp is covered in honey."

John wondered what kind of town Snell had lived in before being imprisoned in the candle, and whether ramps with honey on them were a normal thing there.

"What's happening?" Snell asked.

I'm rethinking my life choices, was what John wanted to say, but he didn't. Instead, he tried to pretend. He felt his right side sticking in the honey, felt it grabbing onto him as he slid. And he felt himself turn towards the honey side.

"I'm turning," he said.

"Exactly," Snell said. "Usually, I use a broom as an example, but I thought this might be more relatable."

John blinked. "So . . . you thought a honey slide would be the most relatable thing for me?"

Snell shrugged. "I don't know a lot about kings."

John decided to let this slide. Then, he giggled at his own pun, but only internally as he didn't want to offend Snell.

"Light turns when it goes from one medium to another. If it goes into something where it goes slower, it turns to go deeper into the medium. If it goes into a medium where it's faster, it leaves at a shallower angle."

"Ok."

"So, make me a rainbow."

"What?"

"Turn yourself into a beam of light and then make yourself into a rainbow. That is your task."

"But . . ."

Snell moved to the far side of the tower and crouched down out of the wind. A transparent deck of cards appeared in his hands, and he began playing solitaire.

"Okay," John muttered to himself. "I'm light. I am made up of all the colors. So . . ." he raised his voice and shouted at Snell, "is it a trick question? Am I already a rainbow?"

Snell gave him a look and turned back to his cards.

Darn. "Also, I turn when I go from a place where I'm fast to a place where I'm slow. How does that help? If I turn, all of me must turn." He sighed. "Ok, different idea. When are there rainbows? When it's rainy. Ok, well, that makes sense if water makes me turn."

John leaned his elbows on the wall and propped his forehead on his hands. The stones were hard and cold under his elbows. "But just shining light into water doesn't make a rainbow. Some stones also make rainbows. Do I have any of those?"

Excitedly, John jumped up and ran down the steps. He bumped into a soldier who said something to him, but John was too excited. Strangely, he could hear the soldier jogging after him as he ran towards the royal vault. When he reached the door, he had to fumble around for his keys, giving the soldier time to catch up.

"Sir, I'm sorry to interrupt, sir—"

"Oh, don't worry, it's fine," John said, trying a ruby key in the lock. No good. He frowned. Was it the emerald one?

"No, sir, it's not just that; it's the peasants, sir."

"Oh?" No, it was definitely the iron one. A serious looking key for a serious thing like a vault. He slotted it into the keyhole and smiled when it clicked into place.

"They're . . . they're attacking the castle, sir."

"What?"

"Er, well, they've been . . ."

"Not too happy with me, no. I understand," John said, but his mind was still on rainbows. Were those crystals still here?

"I assume you'd like us to barricade the doors?"

"Yes, yes. Defend the cast—" John looked up. "Wait, did you say the peasants are attacking the castle?"

The soldier sighed in relief. "Yes, sir."

"Why?"

"It's the protesters, sir. They've gotten out of hand. What would you like us to do?"

John stroked his chin, wishing he had a beard. He didn't blame the peasants. He'd been a terrible king. If they could just wait a little longer, he would be a light mage and then he could

fix everything. In the meantime, he didn't want his soldiers killing them. It wasn't their fault.

"Defend the castle, but don't hurt any of them."

"Er, what sir?"

"Can you keep them out without hurting them?"

"Well, maybe, but if they start lighting things on fire or—It looks like they're bringing in a battering ram. The castle doors are solid but they're not impenetrable."

"Well, just hold them off as long as you can, all right?"

"If we can't, sir?"

"Come get me. I'll be on the north tower. Let them in if you have to; tell them I'll be down shortly, but don't hurt any of them, all right?"

"Er . . ."

John realized he might have to be commanding here. "That's an order!"

The man gave a curt nod. "Right. Of course, sir." He hurried off down the corridor.

Glad to be rid of the distraction, John let himself into the vault. There were piles of gold and silver and jewels. Maybe if he gave some of these to the peasants, they'd be less angry, he thought. It wasn't a terrible plan. Although, most of his plans didn't turn out well, so he probably shouldn't have a lot of confidence in this one.

He held up a diamond. Faint glints of color shone through it, almost rainbows but not quite right. He knocked over a pile of gold as he moved to the back of the vault. There at the far end was a pile of grubby crystals. King John the Fourth had been a fortune teller and had liked having what he called 'mystical

stones' around. That was one ancestor John wasn't completely intimidated by.

He held up a crystal and grinned when he saw the rainbow pattern shining through it. He noticed how the light went straight from the candle burning in the wall sconce, through the crystal, and made a sharp bend right as it passed through, splitting into a rainbow.

"Oh, each of the colors bends a different amount," he muttered to himself. "Cool."

He sprinted back up the stairs of the tower, the crystal held tightly in his fist. His leg muscles were burning, and his breath came in gasps. This was already more exercise than King John had had in the last three weeks put together.

When John at last reached the top of the tower, the ghosts were all playing a card game, except Einstein who was pouting in the corner, having lost several hands in a row.

"I've got it!" John shouted. He grabbed the candle, focused on the wick, and immediately shot off into the night, dropping the crystal.

When he appeared back on the rooftop, he was relieved to see the crystal was unharmed. He tried again, this time more carefully. He managed to slow time enough to turn, aligning himself with the crystal, and this time, when he shot off into the darkness, he split into a magnificent rainbow before hurtling off in all directions.

"Correct," was all Snell said.

"Now it's our turn," the dapper ghost said, rubbing his palms together.

The Wave Nature of Light

"I am Louis de Broglie," the dapper one said. He pointed to the ghosts beside him. "This is Thomas Young and James Clerk Maxwell, and we will each show you what you are."

"I thought I was a person," John said.

"We mean when you're light," de Broglie said.

"Aren't I light then?" John asked.

"Yes and no," de Broglie said. "Young Thomas will demonstrate."

"I hate it when you call me that," Thomas Young said.

"But your name lends itself to it so well," de Broglie said.

"Exactly, you think I haven't heard it a million times before?"

De Broglie didn't respond, so Young turned to John.

"You've probably been thinking of yourself as similar to that potato that hit you in the head earlier today," Young began.

"You saw that?"

"Yes," Young said, and John winced. "Anyway, light is much more than a potato."

"I would imagine."

"No, you wouldn't," Snell interjected.

For someone so quiet, he was suddenly very cranky, John thought. Maybe he was hungry. If ghosts could even be hungry.

"For this demonstration, you will need a piece of parchment."

After another long jog down the tower and back up again, John presented Young with the parchment.

"Excellent. Now cut two slits in it, extremely close together."

John stared at him for a few seconds, then, grudgingly, ran back down the stairs, got a knife, and ran back up again. He cut two slits in the paper, about an inch apart.

"I said extremely close together!" Young said. "So close you can barely tell they are two cuts."

It took almost fifteen tries for John to get it right.

When at last Young was satisfied, he addressed John. "Now, if you were a potato, a very tiny potato, and you were thrown at these slits, what would happen?"

"I'd go through one of them."

"Exactly."

John waited for him to elaborate.

Young smiled in a very self-satisfied way. "Now, if you travel towards these slits as a beam of light, what do you think will happen?"

"Er, I'll go through one of them?"

"See! You are thinking of yourself as a potato!"

John didn't think that logically followed.

"Try it," Young said.

Obligingly, John took the candle, which was beginning to look significantly shorter. Again, he concentrated on the wick, and again he found himself a glowing orb of light. He slowed time and directed himself at the slits that Young held aloft.

John approached the slits; ten feet away, then five, then two, then only a few inches. When he hit the slits, though, a very strange thing happened. He passed through both at once. And that wasn't all. Strangely, he felt himself overlapping. He

was in many places at once, every place at once, and he found that there were parts of himself that, when they overlapped, made more of himself, and other places that, when he overlapped with himself, he disappeared.

This made no sense at all to him, so when he reappeared in human form, he looked suspiciously at the three ghosts in front of him.

"What was that?"

"See, definitely not a potato," Young said.

John felt he'd have to agree with that.

"Although technically potatoes can do that, too," de Broglie said.

"Don't confuse him," Young snapped back.

De Broglie lifted his hands. "I am simply attempting to be accurate."

"Well, you'll have your turn. We'll get there." Young turned to John. "Bring us a bowl of water."

John felt his leg muscles protest. "Is there anything else you'll need? Anything else I should get while I'm down there?"

"I don't think so."

"You're absolutely sure?"

"I can think of nothing else I might require."

John moved slowly to the stairs, glancing back over his shoulder a few times, but none of the ghosts thought of anything or moved to stop him.

Somewhere off in a distant part of the castle he could hear an ominous pounding, but he didn't give it too much thought.

He returned with a large bowl of water, only half of which he'd spilled on himself. Young instructed him make ripples in it with his fingertips.

"See how they expand outwards from your finger when you wiggle it?"

John nodded.

"Light is like that."

"Like water?"

"No, like the wave in water."

"Does it do the same thing but with air, then?"

"No," Maxwell interjected. "That's my bit. It's made of something else. It doesn't need water or air or anything at all to travel through."

"Wait your turn, Maxwell," Young admonished. "Now, John, make ripples with both your fingers at once." John did so. The waves bounced along, and where they overlapped, they made strange, complicated patterns.

"Those patterns," Young said, "are exactly what we see when light passes through two very narrow, very closely spaced slits. Or even though a single narrow slit. That is how we know that light is not like a potato."

Among other ways, John thought. But he was impressed.

"My turn," de Broglie interrupted. "Light is a wave, as is everything else."

"What?" John said, flabbergasted.

"Yes, technically nothing is a potato. Or, just a potato, rather."

This was too much for John, so he just nodded politely.

"Well, that was me. You're up, Maxwell."

Maxwell grinned and stepped up to John, who was still completely confused. “I fear I will only confuse you more, but you asked an important question earlier. You asked if light travelled through air.”

At least he was at the point where he could ask intelligent questions, John thought. That was something.

“What does it travel through?” he asked.

“It doesn’t need anything to travel through. It is its own thing. It is made up of two parts. These two parts change, and in changing, create one another. So, light is first one then the other, and in this way, it travels along through the universe, the fastest thing in the universe.”

“Oh. I see.” He didn’t really, but he doubted it would get clearer.

Just then an enormous crash came from below. John looked down and realized the front gates were on fire and had just collapsed, bashed in by a large battering ram, which was also on fire. The townspeople were now storming in over it, shouting and throwing things. He’d better do something. The soldiers were still under orders not to hurt anyone.

“What is my final test?” John asked. “I need to free you now, before things down there get any worse.”

“Oh. No final test,” Young said.

“What?”

“We didn’t get around to making one up.”

“But you’ve been standing around playing cards all night.”

“You don’t need one. You can burn the candle all the way to the end now.”

“But what if I’m not ready?”

"Then letting us free will kill you."

"Er, isn't there some way for me to practice first?"

"No."

John twiddled his thumbs. Down below, the people were inside the castle now, ransacking and setting things on fire. It seemed unlikely that he could sneak out past them. And he did want to free the five light mages. It was an opportunity to do a real heroic deed. Too good to pass up.

"What do I have to do?"

"You must be stronger than the candle. So, light it, but don't let it transform you. Use what you know about light to keep yourself human."

John didn't give himself time to think. He picked up the box of matches and the grubby, melted end of the candle and lit the wick.

Again, the white, clear flame blossomed on the end, and John felt its pull as it tried to transform him. He sensed the waves of energy coming off it, even with his eyes shut tightly. He found that, even with his eyes closed, he could see everything.

The candle glowed brighter, but John held on, willing himself to stay human. Underneath the light, he began to sense something else. A darkness. Two dark shapes, struggling to get out. The candle was burning down. John tipped it slightly to the side so that the wax ran onto the stones below. At last, it was so short that he couldn't hold it anymore and he set it on the cobble stones. The light grew brighter and brighter,

shining through him with a horrible intensity, but he held on to himself.

"I am a potato," he thought.

The candle went out. And the first dark thing crept out from it. It was a shadowy image—his parents, leaning over the edge of his cradle, smiling at himself as a baby. A servant girl came in. She brought them goblets of wine. Wait, he recognized that face. It was Griselda. He wanted to cry out to his parents, to warn them, but they drank the wine and fell to the ground. Griselda gathered up the cups and left.

There were tears in John's eyes. He was in a dark realm; there was nothing to hear or see now. He sensed somewhere the last nub of wax that had been the candle, but that was it. The ghosts were gone. John wanted to run. He wanted to find Griselda, imprison her, have her executed for her treason. Then he would order the soldiers to repel the invasion. He would put down the revolt and take control of the kingdom, as he should have done in the first place.

But he stayed. There was one dark thing left in the candle, and if he didn't release it the mages would remain trapped. He had promised to free them, and he needed to know what the final secret was. He focused his attention on the candle and waited.

The secret that crawled out was old. Many hundreds of years old. It had been trapped in the candle much longer than any of the five ghosts and much, much longer than the previous secret. John wanted to turn away as it came out. Its sightless eyes passed over him; its teeth glinted in the blackness. But John watched and waited. He saw his parents, then their

parents, then their parents, looking just like they did in the portraits on the walls. They flashed one by one before his eyes, all the way back to King John the First. This King John grew younger; the crown disappeared from his forehead. As John watched, he saw his ancestor was not a prince, but an advisor to some other king and queen. He watched his ancestor betray this king and queen, watched the throne be taken from them. But a little daughter remained. She was snuck out of the castle, in the arms of a maid, and hidden away in the forest. This little girl grew up, had a child. That child grew up and had children in turn. Their children had children, all living in a tiny village in a forest at the edge of the kingdom. Until John saw Griselda.

The secret faded away into mist, and light returned. John stood on the top of the tallest tower, swaying. He was dimly aware of the faint sounds of a battle raging far below him.

He turned and saw the ghosts were no longer ghosts but men; powerful mages who stood before him.

"Thank you for freeing us," Young said.

"Yes, you did well," Maxwell said, his moustache twitching.

"Indeed, quite well," de Broglie said.

"You are now truly a light mage," Einstein said.

There was a pause in which Einstein elbowed Snell.

"Good job," Snell muttered.

There was an uncomfortable silence, then Einstein broke in. "Now, would you like our help quelling this rebellion?"

John considered this. He felt the new power he had over the light, and he knew these five powerful mages would help him. He could rule the kingdom, maybe make it better than

any of his ancestors had before him. But then he thought of Griselda.

Because of her, he had grown up without parents, with no one to teach him how to be a king. But she had also protected him, helped him, taught him things herself. And she, too, had been wronged, many years ago, by his own ancestors. John didn't think he could forgive her, but he knew he was not a very good king. Griselda had been a good leader; she had ruled the kingdom well in his stead. And now he had learned that she was in some ways the rightful queen. It was her family, many generations ago, who were the true rulers of Amrath.

John knew what he had to do.

Without saying anything, he descended the steps of the tower for the last time. He went straight to the throne room, where the soldiers stood in a line, their spears pointed at the villagers, who were attempting to break through.

John held up his hands, and light shone from them. White light shot from his fingers and illuminated every corner of the throne room. The soldiers and villagers alike turned to stare at him, their eyes dazzling. Out of the corner of his eye, John saw Griselda enter the room.

"Good people of Amrath," John began. As they were currently looting and burning the castle, this form of address confused them. "I am sorry. I've been a terrible king. However, I've just

learned that I'm not actually the rightful king. Griselda is. Or queen rather. No need to continue that terrible tradition."

Griselda stared at him, her mouth open. The townspeople weren't sure what to do.

John removed his crown, walked over to Griselda, and placed it on her head. Some of the townspeople put down their weapons. Someone clapped uncertainly.

"Griselda will be a much better queen for you than I was."

There was some more polite, confused clapping. John turned to Griselda.

"Um, your turn. How do we get them to leave?"

Griselda shook herself and straightened her shoulders. "Well, thank you, everyone, for coming," she said. The townspeople looked even more confused. "Tomorrow I'll hold court here in the throne room to hear your grievances. Please come back tomorrow."

There was some grumbling, but this seemed fair. Griselda had always been popular. The peasants picked up their pitchforks and left.

When they were alone, John turned to Griselda.

"I saw what you did," he said, and Griselda looked down at the floor.

"I'm so sorry," she said. "I've wanted to tell you."

At that instant, there was a loud pop, and the sorceress appeared. No longer did she appear as an old woman. Now she was young with a strangely ancient face, glowing with bright purple light.

"Your family have all been terrible for this kingdom. Griselda didn't know what she was doing. I poisoned those goblets

and gave them to her to give to your parents," the sorceress admitted. "I wanted you out of the way, too, but Griselda, when she learned what I'd done, wouldn't hear of it. She protected you. She insisted I give you a chance to learn the truth."

"Is this true?" John asked.

Griselda nodded.

John, relieved that he hadn't just given the kingdom to a murderer, sighed. "Well, good. You'll make a much better king—queen—than I would have. I didn't really want to be king anyway."

"Thank you," Griselda said. "I'm sorry. Even though I didn't know what I was doing. I still have wished every day that I hadn't done it."

"Thank you for giving me the opportunity to learn the truth," John said and felt pleased with himself. That was something a noble knight in a story might have said.

John smiled at Griselda, who smiled back.

Einstein appeared behind them, wheezing. "Those are terrible stairs," he gasped. "No wonder you complained so much."

The other light mages filed in behind him, eyeing the sorceress warily.

"And we were simply descending," de Broglie said. "The ascension must be even more arduous."

"We were stuck in a candle for several hundred years," Snell said, pointing to John. "He's been out walking around."

John laughed. He was no longer King John the Fifteenth, or even Prince John. Now he was just John. Well, John the light mage. Rider of the light. And dragons. If he could find any.

That same evening, John and the sorceress and the light mages helped Griselda take down the fourteen portraits on the wall, revealing behind them a large painting of a nice-looking family. Griselda's ancestors.

From then on, Queen Griselda ruled Amrath wisely and well. The kingdom flourished, and she was known far and wide as the greatest ruler the kingdom had ever had. She had help, though, from her six court mages, who travelled the kingdom, bringing light and truth wherever they went. John moved the portraits of his ancestors to a little-used summer cottage and visited them occasionally. And everyone lived happily ever after.

THE END

THE PHYSICS

The Speed of Light

PART of the inspiration for this story comes from a famous story about Albert Einstein. Einstein was famous for his thought experiments, where he would, without doing any calculations, imagine something, and see what ideas popped out. In one of these thought experiments, he imagined that he was riding a beam of light, and from there all these amazing ideas about time dilation came out. That's how special relativity was born.

But before you get into any of that, you need to know the basics. The first thing you must know is that the speed of light is, as far as we know, the fastest anything in the universe can go. In a vacuum, (meaning in empty space), light travels at approximately 300,000,000 meters per second (m/s), or 186,000 miles per second. We call this speed "c".

The whole circumference of the earth is about 25,000 miles. So, if it were able to curve, light could go all the way around the earth nearly 8 times in one second.

The distance from the Earth to the moon is about 239,000 miles, meaning it takes light less than two seconds to get there.

TRY THIS:

There are a lot of fun things to think about here, in terms of whether the speed of light is the fastest anything can go. Try watching this minute physics video and see what you think of his argument: How to Break the Speed of Light.

Is the dot of the laser pointer really breaking the speed of light? Can you think of any other ways of trying to break the speed of light?

After you've thought about it, watch this Veritasium video: Will This go Faster than Light?

This Vsauce video, too, talks about some of the same ideas, in a really entertaining way: What is The Speed of Dark?

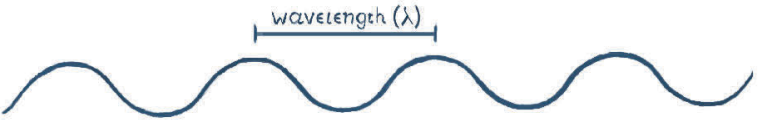
Color

Now let's talk about some of the properties of light. First off, what is color? Well, basically, different colors of light are different energies of light.

We need to talk about some properties of waves at this point. Specifically, frequency and wavelength. I'm going to get into a little math here. You can skip over this part if math isn't your thing or you haven't taken algebra, and the following stuff will still make sense. Having the math just gives a deeper understanding.

Because light is a wave, it has a wavelength. A wave is just a repeating pattern, and the wavelength is the distance that it takes for that pattern to repeat. For example, in this wave:

The wavelength is the distance from one peak to the next:



The wavelength is the distance the wave goes before it starts to repeat the same pattern over again. Notice the symbol in parentheses after the word wavelength. That is the Greek letter lambda, and it is the symbol we use to represent wavelength.

Because light is a wave, it has a wavelength. The wavelength of visible light is so short that it's measured in nanometers (nm). There are 1,000,000,000 nanometers in one meter.

Light also has a frequency. Frequency is just how quickly a wave repeats. It is measured in cycles per second. Does it take one second for the wave to go down and back up again? If so, the frequency is one, because the wave goes through one cycle each second. If it takes half a second for the wave to go down and back up again, then the frequency is two. (Because the wave travels through two cycles in one second.)

In physics, the unit we use to measure frequency is the Hertz (Hz), pronounced "hurts". If a wave has a frequency of 2 Hz, it means that it goes through two cycles every second.

The speed, frequency, and wavelength are all related. This makes sense, because speed is just how much distance we travel in a given amount of time. Speed is how many meters something travels every second. Wavelength is the distance travelled in each cycle, and frequency is how many of those cycles we go through every second. So, if I multiply them together, I get the speed. This gives us this equation:

$$V = \lambda \cdot f$$

Here, v stands for velocity, which is measured in meters per second (m/s), lambda stands for wavelength, which is measured in meters (m), and f stands for frequency, which is measured in Hertz (Hz). So, the speed of a wave is equal to its wavelength multiplied by its frequency.

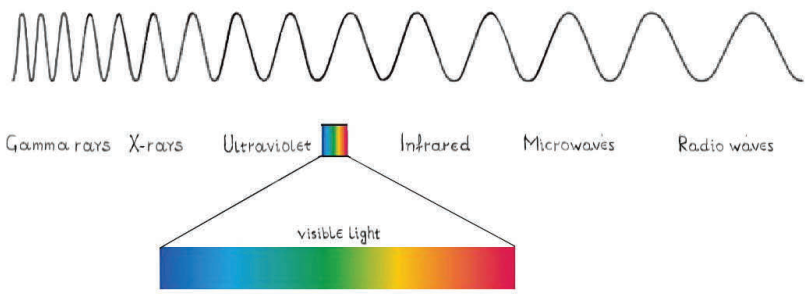
For light waves—and sound waves, and many other types of waves—because the velocity is generally constant, the frequency and wavelength are related. The higher the frequency, the shorter the wavelength. (In math, we would say that wavelength and frequency are 'inversely proportional'.)

OK, now that we've got those math fundamentals out of the way, we can talk about the fun stuff. Different colors of light are different frequencies of light. It's exactly the same as sound. Sound is also a wave (although a different type). With

sound, the higher the frequency, the higher the pitch. The lower the frequency, the lower the pitch.

Similarly, with light, different frequencies/wavelengths give us different colors.

You can see that red light is made up of lower frequencies than blue light. The figure above is called the 'visible spectrum' because it's the wavelengths and colors that we can see. However, this spectrum keeps going. Past the red side is light with even longer wavelengths and lower frequencies. Once we go past red, we get to infrared light, then microwaves, then radio waves. All of these are just light waves that we can't see.



On the other side, past the blue part of the spectrum, we get to light with even shorter wavelengths and higher frequencies. These are ultraviolet light, then x-rays, then gamma rays. You might have heard these called "electromagnetic radiation," but that's just another name for light rays.

The way we see and perceive color is fascinating. First off, white light is a mixture of all the colors of light. Different objects absorb different colors of light. White objects reflect all (or most) of the colors, which is why white or light-colored cars don't get as hot in the sun as dark-colored cars do. Green objects absorb all the wavelengths of light except green. The green light gets reflected, and that's what we see.

So, color is really a process of subtraction. If you see something orange, it means that every color except orange was absorbed by that object.

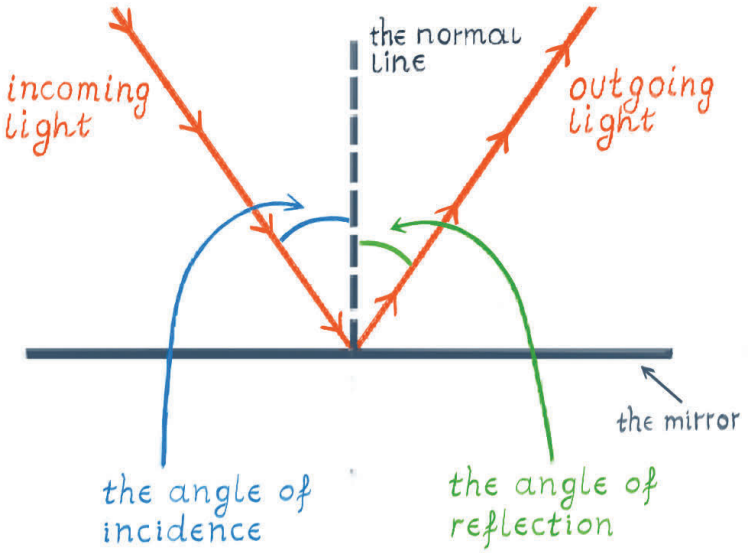
Reflection

Here's a fun brain teaser. How tall of a mirror do you need to have to be able to see all of yourself in it at once? (Meaning, if you're six feet tall, how tall does your mirror need to be? Can it be any size?) I'll tell you how reflection works first, then you can think about it a little before I give you the answer.

So, we already know that light travels in a straight line. In the story, I had John riding the light in straight lines, and then bouncing off various objects. He wasn't totally clear how that was working because it was all happening too fast, but he had a sense that he was sort of glancing off things.

Here's how it works: it depends on what angle you hit something at. If you hit it at a very shallow angle, you'll bounce off at a shallow angle. If you hit it near dead-on, then you'll bounce out the same way, too. If you hit a surface exactly dead on, you'll bounce back exactly the way you came.

The way we say this in physics is "The angle of incidence equals the angle of reflection." The "angle of incidence" is a fancy way of saying the angle you hit something at. The only slightly weird part about this is how we measure the angles. You'd think we'd measure them from the surface, but we don't. We measure the angles from something called a "Normal Line". See the picture below to get a better idea of what I mean by this:

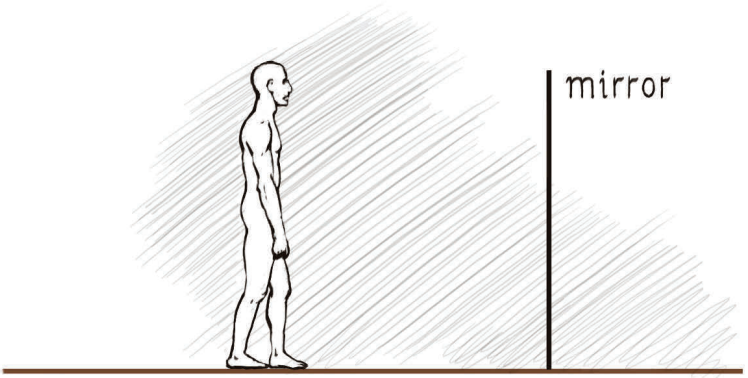


See how the incoming light comes down, hits the mirror, and then bounces off again? The dotted line is the normal line. In math, the word normal means perpendicular (or something that intersects at a right angle, a 90-degree angle). So, a normal line is just a line that is perpendicular to the mirror or surface we're talking about.

Now, the other piece of the puzzle you need to figure out the mirror brainteaser I gave you is that for us to see something, light has to go from the object, bounce off the mirror, and hit our eye.

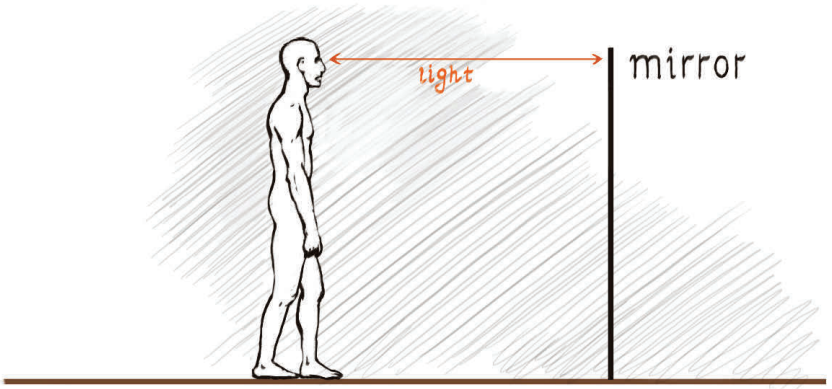
I'll give you the answer on the next page, so think about it for a second before you continue if you want to.

Ok, here's someone standing in front of a mirror:

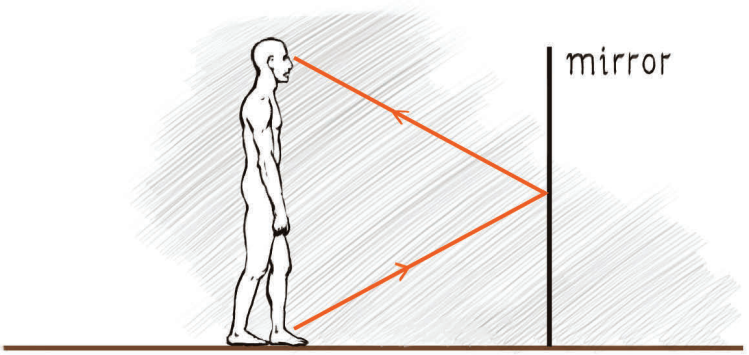


To make this easier, I'm just going to pretend their eyes are basically at the exact top of their head. Now, they can see their eyes by looking straight ahead, right? Light bounces off their

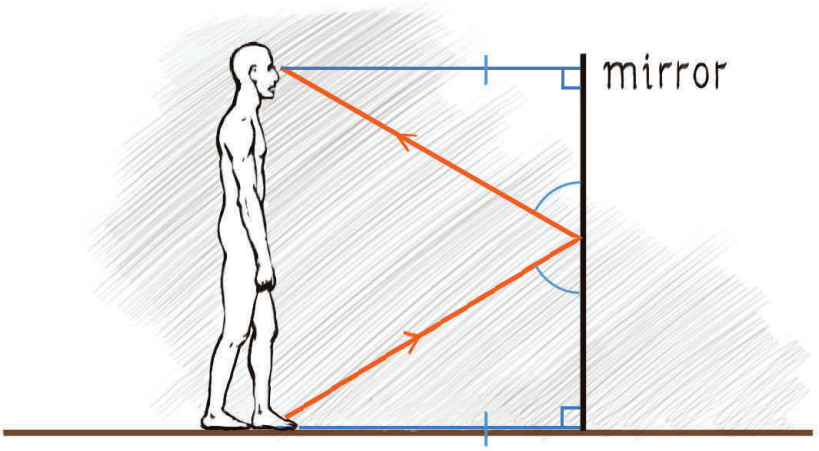
eyes, goes straight to the mirror, then bounces back to their eyes again, like so:



But what about their feet? How do they see their feet? Light has to bounce off their feet, bounce off the mirror, and then bounce into their eyes, like this:

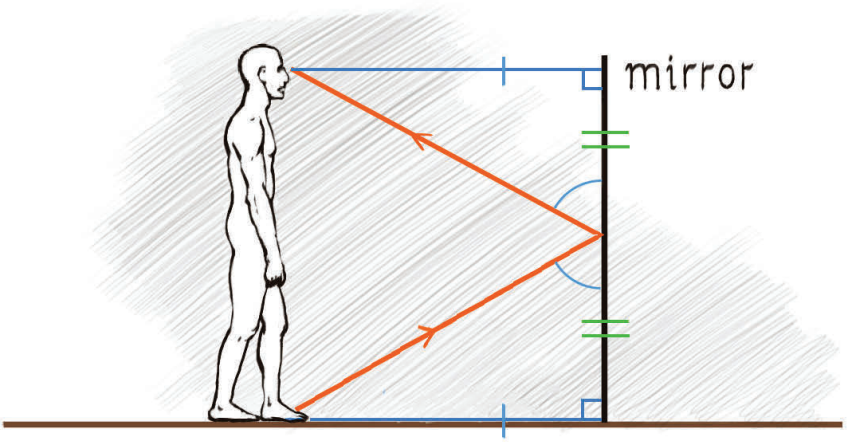


The place where the light from their feet bounces off the mirror is exactly in the midpoint of the mirror. If we wanted to be really fancy, we could prove this with geometry, like this:



(You can ignore this part if you haven't taken geometry.)

Here's the fancy geometry proof, though: the two blue triangles are the same. They each have a ninety-degree angle. Also, the two angles where the light bounces off are the same (because we know that's how light works). Also, this person's feet are the same distance from the mirror as their face is (assuming they're standing up straight). We know that if two triangles have two angles the same and one side that's the same, then they are the exact same shape (we call this being 'congruent' in math). So, therefore, the remaining sides must be the same:



Meaning, the light from their feet hits the mirror in the exact middle. Their shins, by the same argument, are reflected a little higher on the mirror than that. Their arms are somewhere in between their feet and their head.

So, they never need to use the bottom half of the mirror at all. So, the answer to the riddle is that you only need a mirror that is half as tall as you are to see your whole self. This seems crazy, but it's true.

Also, here's another counter-intuitive thing. If you back up while you're looking at a mirror, will you be able to see more of yourself? Think about it, maybe draw some pictures, then go try it.

Snell and the Law of Refraction

Willebrord Snell was born in 1580 in the Netherlands, and he gave us the law of refraction, which tells us how light bends as it goes through different mediums.

In the story, when John is a beam of light, he shoots from the air into the water. He finds that when he hits the water he turns deeper into the water. This bending of the light beam is called "refraction".

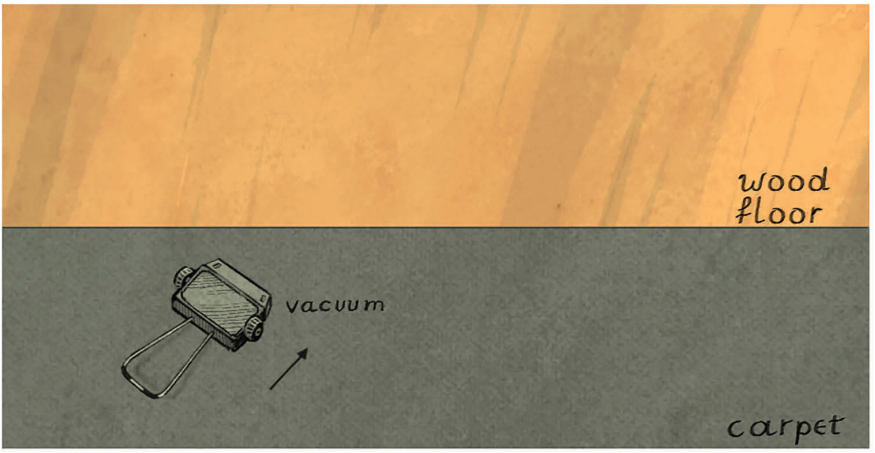
This bending happens because light travels at different speeds in different mediums. For example, when it travels through water, it only travels at approximately 225,563,909 m/s instead if 300,000,000 m/s.

In physics, we keep track of this with something called the "index of refraction" or "n". The speed of light in a vacuum (which we call 'c') divided by the speed of light in the medium, gives us the index of refraction. Here's the equation:

$$\frac{c}{v} = n$$

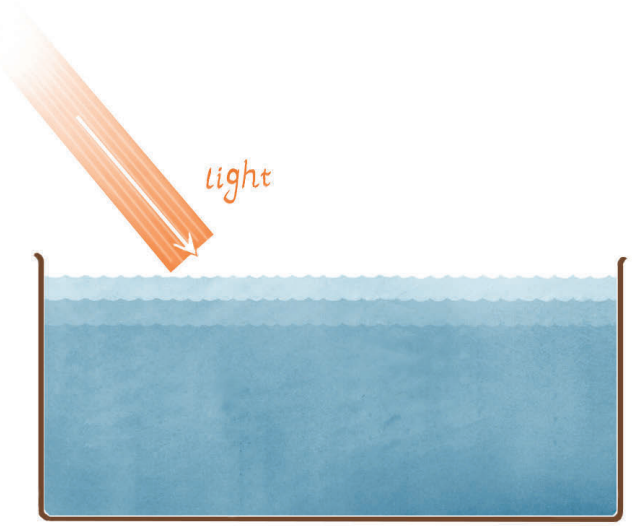
The bigger the value of 'n', the slower the speed of light in that medium. For example, diamond has an index of refraction of about 2.417. Water has an index of refraction of about 1.33. So, light travels more slowly through diamond than through water.

The example I gave in the story for why light bends when it travels though different mediums was the honey on the slide, but more common examples involve lawn mowing or vacuuming. For example, imagine you're vacuuming across a carpet, but there's a hardwood floor or a tile floor or something next to you. Imaging that you push the vacuum onto the floor, like this:



Notice that the left-hand wheel will hit first. The wheels slide more easily on the wood floor than on the carpet, so for that split-second when the left wheel is on the wood floor and the right wheel is on the carpet, the left side will be going faster, so you'll turn to the right.

Light does the exact same thing. Let's take a look at a beam of light entering a pool:



Here, I've drawn the light beam really wide, just to exaggerate the effect. Notice that the bottom side of the light beam is hitting the water before the top side. The light will travel more slowly in the water, so now the bottom side is going to slow down, kind of like it's dragging on the light beam, turning it more deeply into the water.

This refraction is responsible for all kinds of optical illusions. Like, if you put a pencil into a glass of water so that it's half submerged, you'll notice that the top half of the pencil doesn't appear to line up with the bottom half of the pencil.

Snell's Law gives us a way to calculate exactly how much a beam of light will bend upon going into a new medium. Here's the equation:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Where:

n_1 = index of refraction of 1st medium

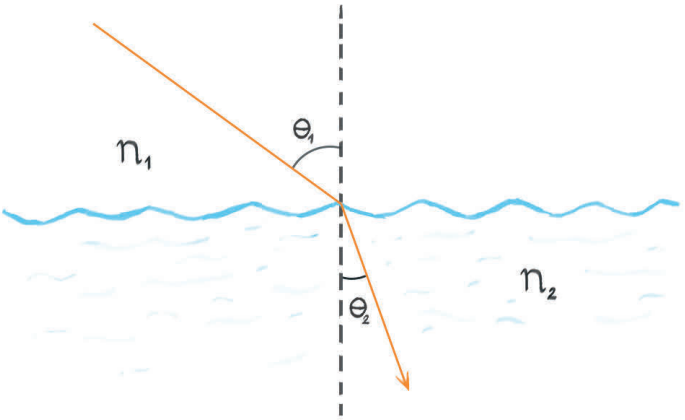
θ_1 = angle of incidence

n_2 = index of refraction of 2nd medium

θ_2 = angle of refraction

You may already know this, but the circle with the line through it is the Greek letter "theta" which we use to represent any angle. Again, we measure all the angles against the normal line.

This equation allows us to calculate the angle that a light beam will travel at when it enters a new medium. See the picture below for a more visual representation of what this equation means:



In this picture, I'm imagining a beam of light travelling from air into water. N_1 would be the index of refraction of air (1), and n_2 would be the index of refraction of water (1.33). If I knew the angle of incidence, I could then use the equation to calculate the angle of refraction.

TRY THIS

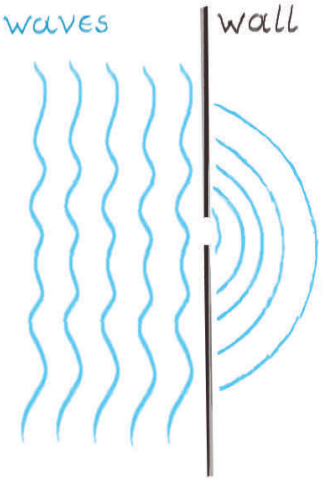
You can get a feel for this by trying out the Phet Simulation from the University of Colorado called Bending Light. Try clicking on the 'prisms' one at the bottom and play around with some of the prisms. Also, try turning on reflections. When light goes into a new medium, some of it goes in but some of it reflects, too. This simulation ignores that unless you turn on reflections.

Electromagnetic Waves

In 1801, Thomas Young performed an extremely famous experiment which is usually today called "Young's Double Slit Experiment". In the story, I had Young demonstrate this for John, and you can try this, too, if you have a laser pointer. Just cut two slits close together (as close as possible; around a millimeter apart) and shine the light at them. What you should see on the other side is something like this:

This was an amazing result, because if light were a particle (or a potato), it wouldn't do this. The pattern should have looked like this:

If light were made up of particles, then the pattern of light would have just been two bright spots, one behind each of the slits. Here's why this happens: as light goes through a slit, it acts just like a water wave going through a hole in a wall. The water waves would move outwards, like ripples on a pond:



This (the way that waves expand out from their source) is called diffraction. Weirdly, light does the same thing. If the slit is narrow enough, the light diffracts outwards as it passes through. If we have two slits, the waves bump into each other as they expand, like this:

Where the waves overlap, we get a complicated pattern of new waves, which is called an 'interference pattern'. The same thing happens with light, which is why we get the alternating bright and dark spots, rather than just two bright spots, and it was evidence for light being more than just a particle.

Even more weirdly, light is not the only thing that does this. All particles do this. Now we're getting into quantum mechanics, but electrons have been demonstrated to do this, too. Technically, all matter can be described as waves, or having wave-like properties.

Einstein contributed to this idea with his paper on the photoelectric effect in 1904, which demonstrated how light can knock electrons loose from metal, and de Broglie is credited with this equation (the de Broglie equation), which gives the relationship between the momentum of any piece of matter and its wavelength:

$$p = \frac{h}{\lambda}$$

Here, p stands for the momentum of a particle, h is something called Planck's Constant, which is 6.6x10^-34, and lambda is the wavelength of the particle. So, you could technically calculate your own wavelength.

Once we knew that light behaved like a wave, people were confused because light travels through space and, as far as we knew, there wasn't anything out there to travel through. All other waves we know of, like water waves or sound waves or vibrations in the ground, they all need some medium to travel through. This is why in space you can't hear anything. There's no air for the sound to travel through. But light from the sun gets to us just fine.

Initially, in the late 1800's, people guessed that there must be something out there for light to travel through. They called this the "luminiferous aether", which is an awesome name. (Also, it's sometimes spelled 'ether'.) They thought that maybe space was full of this stuff, and that light was propagating through it.

However, in 1887 a very famous experiment, the Michelson-Morely experiment, showed that this was not the case.

(Basically, because the earth is shooting around the sun, we should be travelling really quickly through this magical ether. It would be kind of like sitting in the middle of a river with the water flowing around you. Boats travelling upriver should be slower than boats travelling across the river. Similarly, light flowing in the opposite direction as the ether should have been slower than light travelling with the ether or across the current. Michelson and Morely basically set up a race between two light beams, one travelling with the expected "current" and one travelling across the "current," and they tied.)

So, if light wasn't travelling through ether, how was it getting through the vacuum of space? Well, the answer is pretty insane, and it has to do with electricity and magnets.

Magnets have something around them called a magnetic field. You can see the effects of this field when magnets attract pieces of metal. Also, charges, like static electricity, have fields around them. You can see this when a sock sticks to another sock after coming out of the dryer. The socks are charged, and so they attract (or repel, depending on the charge).

In 1865, James Clerk Maxwell published a paper entitled "A Dynamical Theory of the Electromagnetic Field". In this paper, he described one of the most amazing physics discoveries ever made. (Maxwell is often described as one of the greatest physicists of all time, ranked after only Newton and Einstein.) These are his equations:

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times B = \mu_0 \left(J + \varepsilon_0 \frac{\partial E}{\partial t} \right)$$

You may have seen people wearing t-shirts with these equations on them. This is because they're amazing. You need to have taken Calculus to understand them, and that's beyond the scope of what I want to cover here, but basically, what they mean is that if you change a magnetic field, that change creates an electric field. This is how hydroelectric power works. And wind power. Both of these just use forces in nature to turn a turbine. The spinning part usually has a magnet of some kind on it inside a coil of wires. The spinning magnet creates a changing magnetic field. This changing magnetic field creates an electric field, which pushes electrons (charges) around, which creates current.

But the fun doesn't end there. He also demonstrated that a changing electric field also creates a magnetic field. This is how electromagnets work. And how those old Morse code tapping devices worked. If you put a current through a coil of wire, and then change that current, a magnetic field gets created. The magnetic field acts exactly like there was a magnet there inside the coil of wire.

OK, now for the really insane part. Light is basically just those two things combined. A changing magnetic field creates a changing electric field which creates a changing magnetic field. And so on and so on forever. And it's the fastest thing in the universe, and it travels billions of miles through the vacuum of space, totally unassisted by any medium. And we use it to see.

The speed of light can be calculated using constants that were derived from electric and magnetic equations. I don't know how amazing anyone else thinks that is, but I was totally blown away when I learned that.

That is why light is called an electromagnetic wave. Because it is a wave that is made up of changing electric and magnetic fields.

And that's about it. Light is an electromagnetic wave which propagates incredibly quickly. The different colors we see are the different energies/wavelengths/frequencies of light, and there are lot more that we can't see. Radio waves are also light, as are x-rays and microwaves. All of these light rays travel in straight lines, but they slow down when they enter different mediums, which causes them to bend or reflect.

The constancy of the speed of light is also the jumping off point for the theory of relativity, and the wave nature of light is a jumping off point for quantum physics, both of which I'll tackle in other fairy tales.

THANKS FOR READING!

I hope you enjoyed this introduction to the physics of light! If you have any questions, you can email me at SarahAllenPhysics@gmail.com.

You can find free physics worksheets with answer keys on my website, www.MathwithSarah.com.

The next section includes some conceptual practice problems for you to think about, if you're interested. Enjoy!

Conceptual Practice Questions

1. If you shine red light onto a blue piece of paper, what color will the piece of paper appear to be?
2. Why does a mirror swap left and right but not up and down? (Meaning, if you look at yourself in the mirror, your right hand becomes your mirror image's left hand.)
3. If you see an image of a person reflected off a car window, how can you tell where they are?
4. Why can we see through glass?
5. Why is it hard to tell exactly where something underwater is, if you're looking at it from above the water?
6. When light travels from diamond into air, does it bend further from the normal, or closer to the normal?
7. If you put a diamond underwater, does light bend more or less when going from the water into the diamond, compared to how much it bent when it went from air into the diamond?
8. What are some examples of waves in everyday life?

Conceptual Questions Answer Key

1. It will appear black. The blue paper reflects blue light and absorbs red light. So, if you only shine red light on it all the light will be absorbed, and it will appear black.
2. This is a little bit of a trick question. It is more about the way the terms left and right work, rather than anything about mirrors. The mirror doesn't really swap side-to-side. Your hand closer to the right side of the mirror is still closer to what (from your perspective) is the right side of the mirror. What's happened is that the image of yourself in front of you has been turned around. So, from your image's perspective, what was your right hand is now their left hand.
3. Imagine tracing a straight line from your eyes straight to where you see the image on the window. Then, based on the angle that the surface is at, remembering that the angle it comes in at has to be the same as the angle it bounces off at, imagine tracing a line in the new direction.
4. We can see through glass because light travels through it.
5. It's hard to locate things underwater if you're above the water because the light leaving the water bends. Our eyes assume that the light went in a straight line (because that's what it usually does) so we think it came from a different place than it actually did.

6. When light travels from diamond into air, it is travelling from a slower medium to a faster medium, so it bends away from the normal.
7. When you put the diamond underwater, the light bends less because the speed difference between water and diamond is less than the speed difference between air and diamond.
8. Water waves, sound waves, vibrations through floors/walls. Also, if you take a jump rope and flick it and watch the bump travel along the rope, that's a wave. Earthquakes are waves travelling through the ground.